Coding & Solution

Date	06 July 2026
Team ID	SWTID-2026-4324
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Nidhrabingi-Kavitha/OptiCrop

Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy, Bootstrap
Key Features Implemented	• Crop recommendation using Machine Learning. • Soil and weather data analysis. • Yield prediction. • User-friendly web interface. • Fertilizer recommendation. • Data visualization and report generation.
Pending / Incomplete Features	• Real-time IoT sensor integration. • Weather API integration. • Mobile application. • Multi-language support. • Cloud deployment and advanced analytics.
Setup / Run Instructions	1. Install Python 3.10+. 2. Install required libraries using pip install -r requirements.txt. 3. Run the Flask application using python app.py. 4. Open http://127.0.0.1:5000 in a web browser. 5. Enter crop and soil details to use the system.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
		Yes

6	Code is committed to a version control repository	Yes
5	The application runs without critical errors	

Additional Notes / Comments:

The OptiCrop system was developed using Python, Flask, and Machine Learning techniques to optimize agricultural production. The application provides crop recommendations, yield prediction, and fertilizer suggestions based on user input. The code is modular, welldocumented, and designed for future enhancements such as IoT integration, weather API support, and cloud deployment